Immutable

The combination of globally *distributed* computers that can *cryptographically* verify transactions and the building of Bitcoin's blockchain leads to an *immutable* database, meaning the computers building Bitcoin's blockchain can only do so in an *append only* fashion. *Append only* means that information can only be added to Bitcoin's blockchain over time but cannot be deleted—an audit trail etched in digital granite. Once information is confirmed in Bitcoin's blockchain, it's permanent and cannot be erased. Immutability is a rare feature in a digital world where things can easily be erased, and it will likely become an increasingly valuable attribute for Bitcoin over time.

Proof-of-Work

While the previous three attributes are valuable, none of them is inherently new. *Proof-of-work* (PoW) ties together the concepts of a *distributed*, *cryptographic*, and *immutable* database, and is how the distributed computers agree on which group of transactions will be appended to Bitcoin's blockchain next. Put another way, PoW specifically deals with how transactions are grouped in blocks, and how those blocks are chained together, to make Bitcoin's blockchain.

The computers—or miners as they're called—use PoW to compete with one another to get the privilege to add blocks of transactions to Bitcoin's blockchain, which is how transactions are confirmed. Each time miners add a block, they get paid in